

Aaron Chou

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EDUCATION

- National Taiwan University (NTU)**, *B.Sc. in Mechanical Engineering* Taipei, Taiwan
• Last 60 GPA: 4.03/4.3 | CGPA: 3.80/4.3 Sept. 2018 – Jan. 2023
• **Coursework:** Automatic Control, Digital Control System, Kinematics, Dynamics, Computer Programming
- Aoyama Gakuin University (AGU)**, *Exchange Program* Tokyo, Japan
• **Coursework:** Data Structures and Algorithms, Introduction to Computer Systems Sept. 2022 – Jan. 2023

PUBLICATIONS

[1] **Y.L. Chou**, Y.S. Luo, L.C. Wang, H. Mandala, G.H.G. Christmann, C.Y. Lee, W.C. Chen, et al. *Feasibility-Guided Planning over Multi-Specialized Locomotion Policies*, IEEE International Conference on Robotics and Automation (ICRA), 2026. ([link](#))

EXPERIENCE

- Inventec Corporation**, AI Center, Robotics, Advisor: Dr. Wei-Chao Chen Taipei, Taiwan
Robotics Research Engineer — Quadruped Robots Mar. 2025 – Present
- **Vision-Language Navigation (VLN)**: Research **VLN** ([link](#)) in unseen environment through natural language instruction.
 - **Learning-Based Control**: Developed a low-level control SDK for quadruped platforms with no official API, enabling deployment ([link](#)) of learning-based locomotion policies trained in **IsaacGym** and establishing reusable infrastructure.
 - **Terrain-Aware Navigation**: Trained a 15 cm gap-traversal policy, implemented a feasibility-guided planning framework supporting skill selection on hybrid terrain, and contributed to an ICRA 2026 submission [1].
 - **SLAM Optimization**: Improved single-LiDAR SLAM by **dual-LiDAR fusion** ([link](#)) in ICRA 2025 QRC, achieving a **65%** reduction in mapping-to-navigation time and enabling rapid terrain reconstruction in dynamic environments.
- Flytech Technology Co. Ltd.** Taipei, Taiwan
Robotics Engineer — Autonomous Mobile Robots Mar. 2024 – Jan. 2025
- Engineered a precise docking system ([demo](#)) by integrating motion control, localization, and a customized rack tracker, achieving navigation accuracy of $\pm 1\text{--}1.5\text{ cm}$ to minimize redocking attempts in production deployments.
 - Developed and maintained **10+ ROS packages** and delivered a patrol demo showcased at Computex 2024.
- Chien Kuo High School**, Robotics team, FRC#8020 Cyberpunk Taipei, Taiwan
Youth Mentor — Mechanical Design Feb. 2022 – Jul. 2023
- Mentored 30+ students for robot design, CAD/CAM, CNC operation, and testing of the competition-ready robot ([demo](#)).

PROJECTS

- Isaac_MoveIt Manipulator Integration**, *Collaborative Project* ([demo](#)) Jun. 2025 – Present
- Integrated **IsaacSim** with **MoveIt2**, enabling manipulator control in simulation for future dexterous research.
- Differential-Wheeled Robot**, *Independent Project* Mar. 2022 – Aug. 2022
- Developed a SLAM-capable mobile robot by integrating LiDAR with an edge-computing platform for real-time mapping with affordable cost under **120\$**, laying the groundwork for later AMR development and quadruped research.

SKILLS

Robotics: IsaacSim/Lab, ROS/ROS2 (Nav2, MoveIt2, FAST-LIO), Path Planning, Sensor Fusion, SLAM, 3D reconstruction (3DGS)
Hardware & Software: Unitree Go1/A1, AMR, Livox LiDAR, RealSense, IMU, RTX 5090; Docker, Git, Blender, PCL, SolidWorks
Programming: Python, C++, Shell Scripting (Bash), LaTeX

HONORS & COMPETITIONS

- 2025 ICRA Quadruped Robot Challenge (QRC) — Participation Certificate, Autonomous** ([demo](#)) Atlanta, GA
2022 FIRST Robotics Competition (FRC) Sacramento Regional, Finalist (Team Mentor) Sacramento, CA
2020 Dean's List Award (top 5% of the class in the semester) Taipei, Taiwan